

a Controlled graph / loss ablation: Top1FDE (m)

Model	Graph	Loss	Overall	Vehicle	Pedestrian	Bicycle
NG-A	No	SoftCE	2.3809	2.7488	1.3471	1.7499
NG-C	No	Error-Aware	2.2547	2.5871	1.3176	1.7499
G-A	Yes	SoftCE	2.3841	2.7511	1.3527	1.7499
G-C	Yes	Error-Aware	2.1995	2.5163	1.3050	1.7499

b Four registered Overall contrasts: first model minus second

Paired contrast	Δ (m)	Adjusted 98.75% CI	Fold1 Δ	Fold2 Δ	Fold3 Δ
NG-C-NG-A	-0.12621	[-0.15369, -0.10114]	-0.08203	-0.18386	-0.11369
G-A-NG-A	+0.00312	[-0.03029, +0.03351]	+0.03632	-0.02159	-0.00409
G-C-NG-C	-0.05522	[-0.07828, -0.03758]	-0.05937	-0.04474	-0.06092
G-C-G-A	-0.18455	[-0.21720, -0.15225]	-0.17772	-0.20702	-0.17051

c All prespecified current vehicle states: Top1FDE (m)

Model	MovingVehicle (n=41,728)	StoppedVehicle (n=23,044)	ParkedVehicle (n=121,939)
NG-A	10.89942	1.25536	0.22877
NG-C	10.20820	1.25205	0.21469
G-A	10.90527	1.23852	0.23517
G-C	9.90648	1.23694	0.21674

ParkedVehicle G-C – NG-C=+0.00205 m; exploratory 95% CI [-0.00019, +0.00448].

nuScenes HeadTrain630 | 3-fold internal ranking OOF | n=260,151 actor-windows

Vehicle n=191,026; Pedestrian n=66,145; MovingVehicle n=41,728. Bicycle: identical frozen fold-R2 route. Paired Overall intervals: family4 Bonferroni.

Frozen predictor trained on overlapping scenes; not an independent official test.